

Behzad Ousat

AI Engineer | LLM Agents & ML | Ph.D. Candidate

✉ behzad.ost@gmail.com
in linkedin.com/in/behzad-ousat
github.com/behzad-ost

📞 786-824-6705
🌐 behzad-ost.github.io
📍 Miami, FL - Open to relocate

AI engineer building LLM agents, model evaluation harnesses, and large-scale data collection systems end to end. Ph.D. researcher in applied ML with peer-reviewed publications in top security and web venues, plus 4 years operating production infrastructure and data pipelines for a high-traffic ride-hailing platform.

EXPERIENCE

Florida International University - Research Assistant, Applied ML | Miami, FL 01/2022 – Present

- Developed **LLM-powered browser agents** that mimic human interaction patterns, reaching 90%+ success on real-world web crawling tasks and generating training data for downstream detection models.
- Benchmarked 10 LLMs for automated vulnerability analysis in **CI/CD** pipelines, building an evaluation harness that measured precision, recall, and latency on patched and vulnerable packages (RAID '26).
- Trained **LSTM** and **Transformer** models in **PyTorch** on browser interaction time series to classify automated agents at 95% accuracy, then quantified false-positive rates for production use.
- Operated **multi-modal** collection pipelines that captured 1.6B+ artifacts (DOM, screenshots, event streams) into **NoSQL** storage, and ran distributed training workflows over the resulting terabytes.
- Measured Captcha deployment across 1M+ websites and built an **object detection** pipeline that solved 12 schemes at 80%+ accuracy.

Tapsi - Software and DevOps Engineer | Tehran 12/2017 – 10/2021

- Created **internal platform tooling** (Python, Bash) that let developers run a single microservice locally against shared **Kubernetes** staging environments, cutting local resource needs by 80%.
- Developed **Node.js** and Python message-passing libraries over **gRPC** and **RabbitMQ** that sustained 10K events per second, instrumented with **Prometheus** and **Grafana**.
- Migrated 60+ microservices to **Kubernetes** and built production data pipelines with **Kafka** and **Apache Spark** that fed Python ML microservices and analytics workflows.

PROJECTS

LLM-Powered Chat-with-PDF System - Startup MVP 06/2025 – 12/2025

- Built a **Chat-with-PDF** system across 30K+ clinical trial documents, using **RAG** and **prompt chaining** on **Google Cloud** APIs to extract inclusion/exclusion criteria and return matches in minutes.
- Implemented ingestion, chunking, embedding, and retrieval pipelines in **Python** on AWS S3, and provisioned the platform with **Terraform** (Elastic Beanstalk, IAM, CloudFront) and **GitHub Actions** CI/CD.

SKILLS

Languages: Python | Node.js | TypeScript | Bash

LLM & GenAI: LLM Agents | RAG | Prompt Chaining | Model Evaluation | Ollama

LLM Platforms: OpenAI API | Vertex AI | Hugging Face

ML: PyTorch | Transformers | TensorFlow | scikit-learn | Pandas | NumPy

Retrieval & Data: Pinecone | PostgreSQL | MongoDB | Redis | Kafka | PySpark

Infra & Cloud: Docker | Kubernetes | Terraform | GitHub Actions | AWS | GCP

EDUCATION

Ph.D. Computer Science, Florida International University 01/2022 – 12/2026

M.Sc. Computer Engineering, Sharif University of Technology 08/2018 – 08/2021

B.Sc. Computer Engineering, University of Tehran 08/2014 – 08/2018

PUBLICATIONS AND SERVICE

Publications: Peer-reviewed papers at **ACM WWW**, **RAID**, **DIMVA**, **ASIACCS**, and the Journal of Information Security and Applications.

Service: Reviewer for IEEE TIFS, RAID, DIMVA, and MadWeb.